

```
In [1]: import nltk
```

```
In [3]: #nltk.download()  
# if you are already downloaded do not run again  
  
#----- if we want to install a specific package  
import nltk  
nltk.download('punkt')
```

```
[nltk_data] Downloading package punkt to  
[nltk_data] C:\Users\omkar\AppData\Roaming\nltk_data...  
[nltk_data] Package punkt is already up-to-date!
```

```
Out[3]: True
```

```
In [1]: filename='metamorphosis clean.txt'  
file=open(filename,'rt')  
text=file.read()
```

```
In [19]: #print(text)  
# we need to clean the data  
# we need to divide into sentences or words  
# we need to remove the spl charcters  
# we need to convert every word to lower case  
# we need to remove some stop words  
# a,an,the no importance  
# extra space we want only alphabets,numericals etc
```

Step-1: Tokenization

- Token means a piece of text that can be able to read
- Token can be a word or a punctuation also can be token
- Generally in NLP task we will remove punctuations
- But in LLM model tasks we will not remove

```
In [4]: from nltk.tokenize import word_tokenize,sent_tokenize  
# word_tokenize will divide into words  
# sent_tokenize will divide into senetence  
# we can use string.split() ==== NLTK framework
```

```
In [6]: words=word_tokenize(text) # text.split()  
words[:20]
```

```
Out[6]: ['One',
        'morning',
        ',',
        'when',
        'Gregor',
        'Samsa',
        'woke',
        'from',
        'troubled',
        'dreams',
        ',',
        'he',
        'found',
        'himself',
        'transformed',
        'in',
        'his',
        'bed',
        'into',
        'a']
```

```
In [9]: # if we do with basic string methods
        # it will not seperate the words and punctuation
        # so better go with tokenization
        text.split()[5]
```

```
Out[9]: ['One', 'morning,', 'when', 'Gregor', 'Samsa']
```

Step-2: Remove the punctuations

```
In [11]: import string
        string.punctuation
```

```
Out[11]: '!"#$%&'()*+,-./:;<=>?@[\\]^_`{|}~'
```

```
In [13]: import re
        pattern=f"[{re.escape(string.punctuation)}]"
        print(pattern)
        re.sub(pattern, '', "hello !")
```

```
[!"#$%&'()*+,-./:;<=>?@[\\]^_`{|}~]
```

```
Out[13]: 'hello '
```

```
In [17]: re.sub(pattern, '', "hello !%# ho> @re you")
```

```
Out[17]: 'hello  ho re you'
```

```
In [21]: import re
        import string
        from nltk.tokenize import word_tokenize, sent_tokenize
        ===== Read the data =====
        filename='metamorphosis clean.txt'
        file=open(filename, 'rt')
        text=file.read()
        ===== Created a Regex pattern =====
        pattern=f"[{re.escape(string.punctuation)}]"
        cleaned_text=re.sub(pattern, '', text)
```

```
#===== Tokenize into words =====  
words=word_tokenize(cleaned_text)
```

Step-3:Lower The words

```
In [38]: words_lower=[word.lower() for word in words]
```

```
In [31]: import re  
import string  
from nltk.tokenize import word_tokenize,sent_tokenize  
#===== Read the data =====  
filename='metamorphosis clean.txt'  
file=open(filename,'rt')  
text=file.read()  
#===== Created a Regex pattern =====  
pattern=f"[{re.escape(string.punctuation)}]"  
cleaned_text=re.sub(pattern','',text)  
#===== Tokenize into words =====  
words=word_tokenize(cleaned_text)  
#===== Lower the words =====  
words_lower=[word.lower() for word in words]
```

```
In [33]: words_lower
```

```
Out[33]: ['one',
'morning',
'when',
'gregor',
'samsa',
'woke',
'from',
'troubled',
'dreams',
'he',
'found',
'himself',
'transformed',
'in',
'his',
'bed',
'into',
'a',
'horrible',
'vermin',
'he',
'lay',
'on',
'his',
'armourlike',
'back',
'and',
'if',
'he',
'lifted',
'his',
'head',
'a',
'little',
'he',
'could',
'see',
'his',
'brown',
'belly',
'slightly',
'domed',
'and',
'divided',
'by',
'arches',
'into',
'stiff',
'sections',
'the',
'bedding',
'was',
'hardly',
'able',
'to',
'cover',
'it',
'and',
'seemed',
'ready',
```

'to',
'slide',
'off',
'any',
'moment',
'his',
'many',
'legs',
'pitifully',
'thin',
'compared',
'with',
'the',
'size',
'of',
'the',
'rest',
'of',
'him',
'waved',
'about',
'helplessly',
'as',
'he',
'looked',
'whats',
'happened',
'to',
'me',
'he',
'thought',
'it',
'wasnt',
'a',
'dream',
'his',
'room',
'a',
'proper',
'human',
'room',
'although',
'a',
'little',
'too',
'small',
'lay',
'peacefully',
'between',
'its',
'four',
'familiar',
'walls',
'a',
'collection',
'of',
'textile',
'samples',
'lay',
'spread',

'out',
'on',
'the',
'table',
'samsa',
'was',
'a',
'travelling',
'salesman',
'and',
'above',
'it',
'there',
'hung',
'a',
'picture',
'that',
'he',
'had',
'recently',
'cut',
'out',
'of',
'an',
'illustrated',
'magazine',
'and',
'housed',
'in',
'a',
'nice',
'gilded',
'frame',
'it',
'showed',
'a',
'lady',
'fitted',
'out',
'with',
'a',
'fur',
'hat',
'and',
'fur',
'boa',
'who',
'sat',
'upright',
'raising',
'a',
'heavy',
'fur',
'muff',
'that',
'covered',
'the',
'whole',
'of',
'her',

'lower',
'arm',
'towards',
'the',
'viewer',
'gregor',
'then',
'turned',
'to',
'look',
'out',
'the',
'window',
'at',
'the',
'dull',
'weather',
'drops',
'of',
'rain',
'could',
'be',
'heard',
'hitting',
'the',
'pane',
'which',
'made',
'him',
'feel',
'quite',
'sad',
'how',
'about',
'if',
'i',
'sleep',
'a',
'little',
'bit',
'longer',
'and',
'forget',
'all',
'this',
'nonsense',
'he',
'thought',
'but',
'that',
'was',
'something',
'he',
'was',
'unable',
'to',
'do',
'because',
'he',
'was',

'used',
'to',
'sleeping',
'on',
'his',
'right',
'and',
'in',
'his',
'present',
'state',
'couldnt',
'get',
'into',
'that',
'position',
'however',
'hard',
'he',
'threw',
'himself',
'onto',
'his',
'right',
'he',
'always',
'rolled',
'back',
'to',
'where',
'he',
'was',
'he',
'must',
'have',
'tried',
'it',
'a',
'hundred',
'times',
'shut',
'his',
'eyes',
'so',
'that',
'he',
'wouldnt',
'have',
'to',
'look',
'at',
'the',
'floundering',
'legs',
'and',
'only',
'stopped',
'when',
'he',
'began',

'to',
'feel',
'a',
'mild',
'dull',
'pain',
'there',
'that',
'he',
'had',
'never',
'felt',
'before',
'oh',
'god',
'he',
'thought',
'what',
'a',
'strenuous',
'career',
'it',
'is',
'that',
'ive',
'chosen',
'travelling',
'day',
'in',
'and',
'day',
'out',
'doing',
'business',
'like',
'this',
'takes',
'much',
'more',
'effort',
'than',
'doing',
'your',
'own',
'business',
'at',
'home',
'and',
'on',
'top',
'of',
'that',
'theres',
'the',
'curse',
'of',
'travelling',
'worries',
'about',
'making',

'train',
'connections',
'bad',
'and',
'irregular',
'food',
'contact',
'with',
'different',
'people',
'all',
'the',
'time',
'so',
'that',
'you',
'can',
'never',
'get',
'to',
'know',
'anyone',
'or',
'become',
'friendly',
'with',
'them',
'it',
'can',
'all',
'go',
'to',
'hell',
'he',
'felt',
'a',
'slight',
'itch',
'up',
'on',
'his',
'belly',
'pushed',
'himself',
'slowly',
'up',
'on',
'his',
'back',
'towards',
'the',
'headboard',
'so',
'that',
'he',
'could',
'lift',
'his',
'head',
'better',

'found',
'where',
'the',
'itch',
'was',
'and',
'saw',
'that',
'it',
'was',
'covered',
'with',
'lots',
'of',
'little',
'white',
'spots',
'which',
'he',
'didn't',
'know',
'what',
'to',
'make',
'of',
'and',
'when',
'he',
'tried',
'to',
'feel',
'the',
'place',
'with',
'one',
'of',
'his',
'legs',
'he',
'drew',
'it',
'quickly',
'back',
'because',
'as',
'soon',
'as',
'he',
'touched',
'it',
'he',
'was',
'overcome',
'by',
'a',
'cold',
'shudder',
'he',
'slid',
'back',

'into',
'his',
'former',
'position',
'getting',
'up',
'early',
'all',
'the',
'time',
'he',
'thought',
'it',
'makes',
'you',
'stupid',
'youve',
'got',
'to',
'get',
'enough',
'sleep',
'other',
'travelling',
'salesmen',
'live',
'a',
'life',
'of',
'luxury',
'for',
'instance',
'whenever',
'i',
'go',
'back',
'to',
'the',
'guest',
'house',
'during',
'the',
'morning',
'to',
'copy',
'out',
'the',
'contract',
'these',
'gentlemen',
'are',
'always',
'still',
'sitting',
'there',
'eating',
'their',
'breakfasts',
'i',
'ought',

'to',
'just',
'try',
'that',
'with',
'my',
'boss',
'id',
'get',
'kicked',
'out',
'on',
'the',
'spot',
'but',
'who',
'knows',
'maybe',
'that',
'would',
'be',
'the',
'best',
'thing',
'for',
'me',
'if',
'i',
'didn't',
'have',
'my',
'parents',
'to',
'think',
'about',
'id',
'have',
'given',
'in',
'my',
'notice',
'a',
'long',
'time',
'ago',
'id',
'have',
'gone',
'up',
'to',
'the',
'boss',
'and',
'told',
'him',
'just',
'what',
'i',
'think',
'tell',

'him',
'everything',
'i',
'would',
'let',
'him',
'know',
'just',
'what',
'i',
'feel',
'hed',
'fall',
'right',
'off',
'his',
'desk',
'and',
'its',
'a',
'funny',
'sort',
'of',
'business',
'to',
'be',
'sitting',
'up',
'there',
'at',
'your',
'desk',
'talking',
'down',
'at',
'your',
'subordinates',
'from',
'up',
'there',
'especially',
'when',
'you',
'have',
'to',
'go',
'right',
'up',
'close',
'because',
'the',
'boss',
'is',
'hard',
'of',
'hearing',
'well',
'theres',
'still',
'some',

'hope',
'once',
'ive',
'got',
'the',
'money',
'together',
'to',
'pay',
'off',
'my',
'parents',
'debt',
'to',
'him',
'another',
'five',
'or',
'six',
'years',
'i',
'suppose',
'thats',
'definitely',
'what',
'ill',
'do',
'thats',
'when',
'ill',
'make',
'the',
'big',
'change',
'first',
'of',
'all',
'though',
'ive',
'got',
'to',
'get',
'up',
'my',
'train',
'leaves',
'at',
'five',
'and',
'he',
'looked',
'over',
'at',
'the',
'alarm',
'clock',
'ticking',
'on',
'the',
'chest',

'of',
'drawers',
'god',
'in',
'heaven',
'he',
'thought',
'it',
'was',
'half',
'past',
'six',
'and',
'the',
'hands',
'were',
'quietly',
'moving',
'forwards',
'it',
'was',
'even',
'later',
'than',
'half',
'past',
'more',
'like',
'quarter',
'to',
'seven',
'had',
'the',
'alarm',
'clock',
'not',
'rung',
'he',
'could',
'see',
'from',
'the',
'bed',
'that',
'it',
'had',
'been',
'set',
'for',
'four',
'oclock',
'as',
'it',
'should',
'have',
'been',
'it',
'certainly',
'must',
'have',

'rung',
'yes',
'but',
'was',
'it',
'possible',
'to',
'quietly',
'sleep',
'through',
'that',
'furniturerattling',
'noise',
'true',
'he',
'had',
'not',
'slept',
'peacefully',
'but',
'probably',
'all',
'the',
'more',
'deeply',
'because',
'of',
'that',
'what',
'should',
'he',
'do',
'now',
'the',
'next',
'train',
'went',
'at',
'seven',
'if',
'he',
'were',
'to',
'catch',
'that',
'he',
'would',
'have',
'to',
'rush',
'like',
'mad',
'and',
'the',
'collection',
'of',
'samples',
'was',
'still',
'not',

'packed',
'and',
'he',
'did',
'not',
'at',
'all',
'feel',
'particularly',
'fresh',
'and',
'lively',
'and',
'even',
'if',
'he',
'did',
'catch',
'the',
'train',
'he',
'would',
'not',
'avoid',
'his',
'bosss',
'anger',
'as',
'the',
'office',
'assistant',
'would',
'have',
'been',
'there',
'to',
'see',
'the',
'five',
'oclock',
'train',
'go',
'he',
'would',
'have',
'put',
'in',
'his',
'report',
'about',
'gregors',
'not',
'being',
'there',
'a',
'long',
'time',
'ago',
'the',
'office',

'assistant',
'was',
'the',
'bosss',
'man',
'spineless',
'and',
'with',
'no',
'understanding',
'what',
'about',
'if',
'he',
'reported',
'sick',
'but',
'that',
'would',
'be',
'extremely',
'strained',
'and',
'suspicious',
'as',
'in',
'fifteen',
'years',
'of',
'service',
'gregor',
'had',
'never',
'once',
'yet',
'been',
'ill',
'his',
'boss',
'would',
'certainly',
'come',
'round',
'with',
'the',
'doctor',
'from',
'the',
'medical',
'insurance',
'company',
'accuse',
'his',
'parents',
'of',
'having',
'a',
'lazy',
'son',
'and',

```
'accept',
'the',
'doctors',
'recommendation',
'not',
'to',
'make',
'any',
'claim',
'as',
'the',
'doctor',
'believed',
'that',
'noone',
'was',
'ever',
'ill',
'but',
'that',
'many',
'were',
'workshy',
'and',
'whats',
'more',
'would',
'he',
'have',
'been',
'entirely',
'wrong',
'in',
'this',
'case',
'gregor',
'did',
'in',
'fact',
'apart',
...]
```

Step-4:Alphanumeric words

- we need alphabets numerals
- any spaces still any unwanted characters

```
In [39]: for words in ['', 'hello', '123', 'hello123']:
        if words.isalnum():
            print(words)
```

```
hello
123
hello123
```

```
In [41]: # for words in ['', 'hello']:
        #     if words.isalnum():
        #         print(words)
```

```
words_alpha_num=[word for word in words_lower if word.isalnum()]
```

```
In [72]: #words_alpha_num
```

```
In [ ]: import re
import string
from nltk.tokenize import word_tokenize,sent_tokenize
#===== Read the data =====
filename='metamorphosis clean.txt'
file=open(filename,'rt')
text=file.read()
#===== Created a Regex pattern =====
pattern=f"[{re.escape(string.punctuation)}]"
cleaned_text=re.sub(pattern','',text)
#===== Tokenize into words =====
words=word_tokenize(cleaned_text)
#===== Lower the words =====
words_lower=[word.lower() for word in words]
#===== Alpha numeric words =====
words_alpha_num=[word for word in words_lower if word.isalnum()]
```

Step-5:Stop words

Stop words are extremely common words, like "the," "a," and "is," that are often removed from text during data processing because they carry little substantive meaning on their own

```
In [46]: from nltk.corpus import stopwords
```

```
In [54]: # nltk.download('stopwords')
```

```
In [52]: stopwords.words('hinglish')
```

```
Out[52]: ['a',
          'aadi',
          'aaj',
          'aap',
          'aapne',
          'aata',
          'aati',
          'aaya',
          'aaye',
          'ab',
          'abbe',
          'abbey',
          'abe',
          'abhi',
          'able',
          'about',
          'above',
          'accha',
          'according',
          'accordingly',
          'acha',
          'achcha',
          'across',
          'actually',
          'after',
          'afterwards',
          'again',
          'against',
          'agar',
          'ain',
          'aint',
          "ain't",
          'aisa',
          'aise',
          'aisi',
          'alag',
          'all',
          'allow',
          'allows',
          'almost',
          'alone',
          'along',
          'already',
          'also',
          'although',
          'always',
          'am',
          'among',
          'amongst',
          'an',
          'and',
          'andar',
          'another',
          'any',
          'anybody',
          'anyhow',
          'anyone',
          'anything',
          'anyway',
          'anyways',
```

'anywhere',
'ap',
'apan',
'apart',
'apna',
'apnaa',
'apne',
'apni',
'appear',
'are',
'aren',
'arent',
'aren't',
'around',
'arre',
'as',
'aside',
'ask',
'asking',
'at',
'aur',
'avum',
'aya',
'aye',
'baad',
'baar',
'bad',
'bahut',
'bana',
'banae',
'banai',
'banao',
'banaya',
'banaye',
'banayi',
'banda',
'bande',
'bandi',
'bane',
'bani',
'bas',
'bata',
'batao',
'bc',
'be',
'became',
'because',
'become',
'becomes',
'becoming',
'been',
'before',
'beforehand',
'behind',
'being',
'below',
'beside',
'besides',
'best',
'better',

'between',
'beyond',
'bhai',
'bheetar',
'bhi',
'bhitar',
'bht',
'bilkul',
'bohot',
'bol',
'bola',
'bole',
'boli',
'bolo',
'bolta',
'bolte',
'bolti',
'both',
'brief',
'bro',
'btw',
'but',
'by',
'came',
'can',
'cannot',
'cant',
'can't',
'cause',
'causes',
'certain',
'certainly',
'chahiye',
'chaiye',
'chal',
'chalega',
'chhaiye',
'clearly',
'c'mon',
'com',
'come',
'comes',
'could',
'couldn',
'couldnt',
'couldn't',
'd',
'de',
'dede',
'dega',
'degi',
'dekh',
'dekha',
'dekhe',
'dekhi',
'dekho',
'denge',
'dhang',
'di',
'did',

'didn',
'didn't',
"didn't",
'dijiye',
'diya',
'diyaa',
'diye',
'diyo',
'do',
'does',
'doesn',
'doesnt',
"doesn't",
'doing',
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```

In [54]: stopwords.fileids()

```
Out[54]: ['albanian',
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```

```
In [56]: #words_alpha_num
# if a word is stop word dont consider that
# for word in words_alpha_num:
#     if word not in stopwords.words('English'):
#         print(word)
eng_stop=stopwords.words('English')
tokens=[word for word in words_alpha_num if word not in eng_stop]
#tokens
```

```
In [58]: import re
import string
from nltk.tokenize import word_tokenize,sent_tokenize
#===== Read the data =====
filename='metamorphosis clean.txt'
file=open(filename,'rt')
text=file.read()
#===== Created a Regex pattern =====
pattern=f"[{re.escape(string.punctuation)}]"
cleaned_text=re.sub(pattern,'',text)
#===== Tokenize into words =====
words=word_tokenize(cleaned_text)
#===== Lower the words =====
words_lower=[word.lower() for word in words]
#===== Stop words =====
eng_stop=stopwords.words('English')
```

```
tokens=[word for word in words_alpha_num if word not in eng_stop]
tokens
```



```
Out[58]: ['one',
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